

DATABASES

Laboratory Work 1

Entity-Relationship Model · International Airport System

The database supports the operations of an international airport: passenger bookings, flight scheduling, airline and airport reference data, boarding, baggage handling, and security screening. Ten entities were identified from the system description; their attributes, keys, and relationships are documented below and visualised in the ER diagram.

1. Entity Identification

- **AIRPORT** — physical airport location
- **AIRLINE** — air carrier operating flights
- **FLIGHT** — scheduled and actual flight record
- **PASSENGER** — traveller information
- **BOOKING** — reservation linking a passenger to a flight
- **BOOKING_CHANGE** — history of changes made to a booking
- **BOARDING_PASS** — boarding document issued for a booking
- **BAGGAGE** — baggage registered for a booking
- **BAGGAGE_CHECK** — inspection result for checked baggage
- **SECURITY_CHECK** — security screening result for a passenger

2. Entities and Attributes

Each table lists the attributes of one entity together with its data type and key role.

AIRPORT

AIRPORT			
Attribute	Data Type	Key	Constraints
airport_id	INT	PK	
airport_name	VARCHAR		NOT NULL
country	VARCHAR		NOT NULL
state	VARCHAR		NOT NULL
city	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

AIRLINE

AIRLINE

Attribute	Data Type	Key	Constraints
airline_id	INT	PK	
airline_code	VARCHAR		UNIQUE, NOT NULL
name	VARCHAR		NOT NULL
country	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

FLIGHT

FLIGHT			
Attribute	Data Type	Key	Constraints
flight_id	INT	PK	
airline_id	INT	FK	→ AIRLINE.airline_id, NOT NULL
departure_airport_id	INT	FK	→ AIRPORT.airport_id, NOT NULL
arrival_airport_id	INT	FK	→ AIRPORT.airport_id, NOT NULL
departing_gate	VARCHAR		
arriving_gate	VARCHAR		
scheduled_departure_time	DATETIME		NOT NULL
scheduled_arrival_time	DATETIME		NOT NULL
actual_departure_time	DATETIME		
actual_arrival_time	DATETIME		
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

PASSENGER

PASSENGER			
Attribute	Data Type	Key	Constraints
passenger_id	INT	PK	
first_name	VARCHAR		NOT NULL
last_name	VARCHAR		NOT NULL
gender	VARCHAR		
date_of_birth	DATE		NOT NULL
country_of_citizenship	VARCHAR		NOT NULL
country_of_residence	VARCHAR		

passport_number	VARCHAR		UNIQUE, NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOOKING

BOOKING			
Attribute	Data Type	Key	Constraints
booking_id	INT	PK	
flight_id	INT	FK	→ FLIGHT.flight_id, NOT NULL
passenger_id	INT	FK	→ PASSENGER.passenger_id, NOT NULL
status	VARCHAR		NOT NULL
booking_platform	VARCHAR		
ticket_price	DECIMAL(10,2)		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOOKING_CHANGE

BOOKING_CHANGE			
Attribute	Data Type	Key	Constraints
booking_change_id	INT	PK	
booking_id	INT	FK	→ BOOKING.booking_id, NOT NULL
change_description	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOARDING_PASS

BOARDING_PASS			
Attribute	Data Type	Key	Constraints
boarding_pass_id	INT	PK	
booking_id	INT	FK	→ BOOKING.booking_id, NOT NULL
seat	VARCHAR		NOT NULL
boarding_time	DATETIME		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BAGGAGE

BAGGAGE			
Attribute	Data Type	Key	Constraints
baggage_id	INT	PK	
booking_id	INT	FK	→ BOOKING.booking_id, NOT NULL
weight_kg	DECIMAL(6,2)		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BAGGAGE_CHECK

BAGGAGE_CHECK			
Attribute	Data Type	Key	Constraints
baggage_checking_id	INT	PK	
booking_id	INT	FK	→ BOOKING.booking_id, NOT NULL
passenger_id	INT	FK	→ PASSENGER.passenger_id, NOT NULL
check_results	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

SECURITY_CHECK

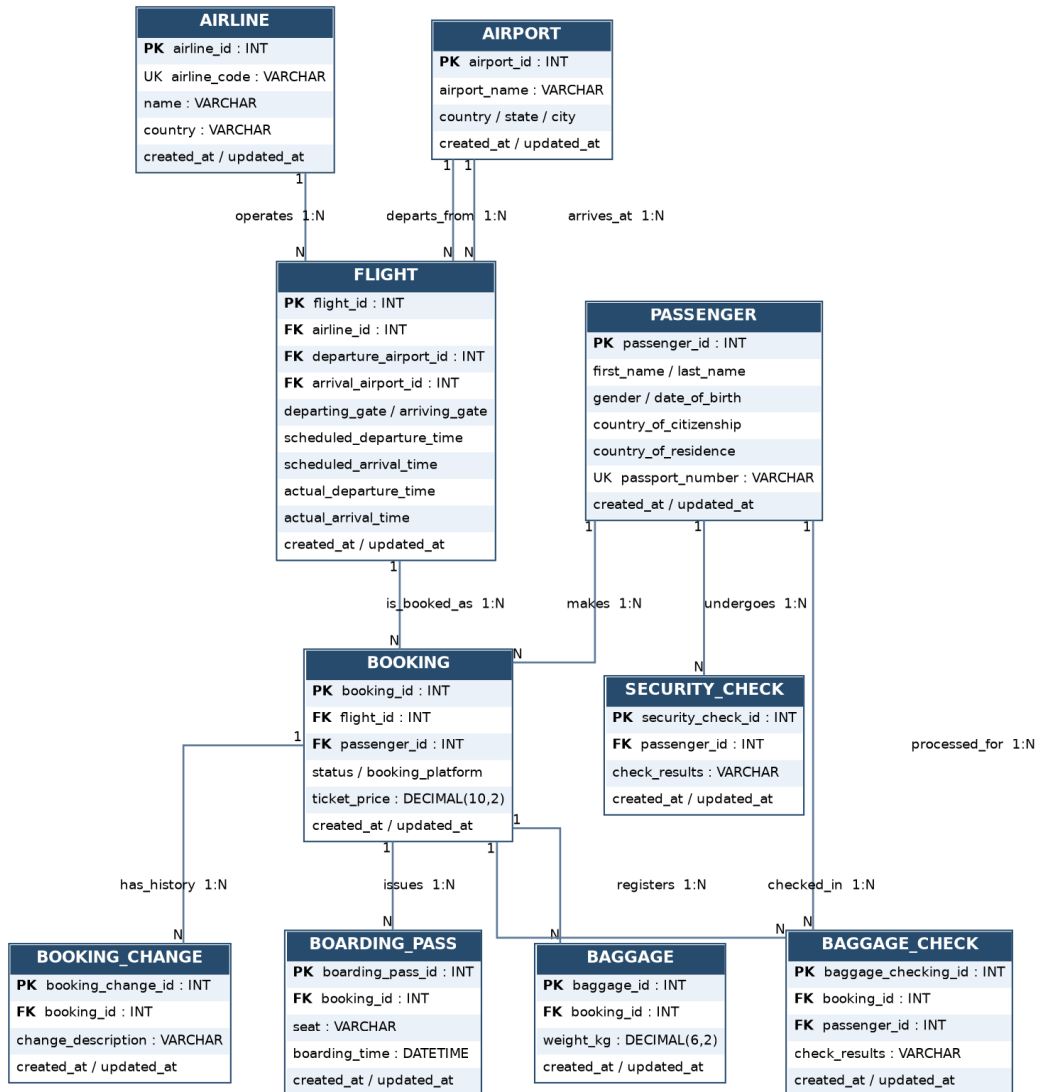
SECURITY_CHECK			
Attribute	Data Type	Key	Constraints
security_check_id	INT	PK	
passenger_id	INT	FK	→ PASSENGER.passenger_id, NOT NULL
check_results	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

3. Relationships and Cardinality

Entity A	Entity B	Cardinality	Meaning
AIRLINE	FLIGHT	1 : N	An airline operates many flights.
AIRPORT	FLIGHT (departure)	1 : N	An airport is the departure point for many flights.

AIRPORT	FLIGHT (arrival)	1 : N	An airport is the arrival point for many flights.
FLIGHT	BOOKING	1 : N	A flight has many bookings.
PASSENGER	BOOKING	1 : N	A passenger can make many bookings.
BOOKING	BOOKING_CHANGE	1 : N	A booking can have many recorded changes.
BOOKING	BOARDING_PASS	1 : N	A booking can issue many boarding passes.
BOOKING	BAGGAGE	1 : N	A booking can register many baggage items.
BOOKING	BAGGAGE_CHECK	1 : N	A booking can have many baggage checks.
PASSENGER	BAGGAGE_CHECK	1 : N	A passenger's baggage can be checked many times.
PASSENGER	SECURITY_CHECK	1 : N	A passenger can undergo many security checks.

4. ER Diagram



5. Legend

Symbol	Meaning
PK	Primary key — the unique identifier of a record.
FK	Foreign key — a reference to the primary key of another entity.
UK	Alternate unique key, e.g. passport_number.
1	"Exactly one" — mandatory participation in the relationship.
N	"Zero or many" — optional, multiple participation in the relationship.